Technical Data Sheet

LOCTITE® LIOFOL LA 3640 / LA 6229

January 2024

Product description
LOCTITE® LIOFOL LA 3640 / LA 6229 provides the following product characteristics:

Technology	Polyurethane
Product type	Laminating adhesive
Condition	Solvent-based
Components	Two components – requires mixing
Application	Film/film and film/foil laminates
Performance	Medium
Cure	Room temperature cure after mixing
Appearance base LOCTITE® LIOFOL LA 3640	In 500 ml glass jar: yellow to brownish, clear to turbid, no precipitation
Appearance hardener LA 6229	Clear, yellowish
Mix ratio, by weight (base:hardener)	15 : 1

Application areas:
LOCTITE® LIOFOL LA 3640 / LA 6229 is recommended for high performance laminates. Besides its high heat resistance, it exhibits excellent chemical resistance against aggressive goods packed in film/film laminates. Laminates manufactured with suitable materials are resistant to retorting, boiling and pasteurising. LOCTITE® LIOFOL LA 3640 / LA 6229 is usually applied at 30 - 35 % solids, exhibiting excellent machinability, good wetting properties and high initial tack.

Suitable films:
- Aluminium foil
- Treated polyethylene
- Treated polypropylene (CPP, OPP)
- Polyester
- Polyamide (CPA, OPA)
- Unplasticized PVC films

Not suitable films:
- EVA-Copolymerfilms

Product properties
The cured polyurethane film is transparent, odourless, elastic and age-resistant.

Technical data
LOCTITE® LIOFOL LA 3640:

Solids content, %	58 - 62
Viscosity, Brookfield - LVT, 20°C, mPa·s	
Spindle 2, speed 12 rpm	210 - 410
Volumetric density, g/cm³	1.05
Component	NCO component
Solvent	Ethylacetate

Technical data
LA 6229:

Solids content, %	100
Viscosity, Brookfield - LVT, 20°C, mPa·s	
Spindle 3, speed 60 rpm	700 - 1 700
Volumetric density, g/cm³	1.17
Component	OH component
Solvent	None

Compliance with food packaging regulations:
Under specific conditions LOCTITE® LIOFOL LA 3640 / LA 6229 is suitable for manufacturing of food packaging materials under the applicable FDA, EC-provisions and/or BfR recommendations.
LOCTITE® LIOFOL LA 3640 / LA 6229 contains aromatic isocyanates. We therefore recommend migration testing of the fully cured material under conditions of use, particularly with high temperature applications, to demonstrate compliance with the limits for primary aromatic amines.
A food contact statement with further details is available on request.
The responsibility for compliance with the specific conditions is with the converter and not with the adhesive supplier.

Directions for use

Preliminary statement:
Prior to use it is necessary to read the **Safety Data Sheet** for information about precautionary measures and safety recommendations. Also, for chemical products exempt from compulsory labeling, the relevant precautions should always be observed. Please also refer to the local safety instructions and contact Henkel for analytical support.

Protective measures:
As the adhesive contains polyisocyanates any contact with skin should be avoided. If skin gets in contact with LOCTITE® LIOFOL LA 3640 / LA 6229 please remove the adhesive immediately. Safety regulations with regard to handling of isocyanates have to be followed as per the guidelines issued by the International Isocyanates Institute.

Dilution:
Suitable solvents are ethyl acetate, acetone and Methylethylketone provided that the water content does not exceed 300 ppm.
Do not use aromatic or alcoholic solvents.

Dilution table
In order to obtain a certain solids content, the following quantities of solvent have to be added.
The Ford/DIN cup (4 mm/20 °C) viscosities are based on an average viscosity of LOCTITE® LIOFOL LA 3640 / LA 6229.

LOCTITE® LIOFOL LA 3640	LA 6229	Ethyl acetate	Solid content	Viscosity (sec.)
25 kg	1.67 kg	6.67 kg	50%	19 - 25
25 kg	1.67 kg	10.37 kg	45%	15 - 19
25 kg	1.67 kg	15.00 kg	40%	12 - 16
25 kg	1.67 kg	20.95 kg	35%	11 - 15
25 kg	1.67 kg	28.89 kg	30%	10 - 14

Mixing
LOCTITE® LIOFOL LA 3610 has to be diluted under stirring with corresponding solvent to the desired application solids. Then the correct amount of hardener LA 6229 has to be added under agitation.

Working life:
The ready-for-use adhesive (30 - 35%) in closed containers has a working time of about 1 day without substantial viscosity increase. Ready-for-use adhesive which has not been consumed in the machine for more than 6 hours has to be disposed of. Otherwise it might lead to unstable adhesive films on the laminates particularly with regard to laminates exposed to high temperatures and filled with aggressive goods thus causing delaminations.

Application:
LOCTITE® LIOFOL LA 3640 / LA 6229 can be applied with all dry laminating units that have smooth rollers. The suitability of gravure cylinders must be checked with long-term tests before running full scale production.

Application weight:
The application weight may range from 2.5 - 3.5 g/m² dry depending on the film or foil combination and the intended use of the laminate. Sterilisable and deep draw laminates require a coating weight of at least 4 g/m². For printed laminates the weight has to be adjusted to higher weights accordingly, it should be approx. 1 - 2 g/m² higher than for unprinted films.

Working conditions:
High laminating temperatures together with a high line pressure of the calander improve the bond strength and optical appearance of the laminates.

Drying:
Drying conditions depend on the efficiency of the dryer and the line speed. As a general rule to avoid over-drying we recommend:

50 - 60°C at a machine speed of 50m/min
60 - 80°C at a machine speed of 150m/min

Curing:
Manufacture of triplex laminates is possible 2 - 4 hours after duplex production. The laminates may be rewound and slit after 1 - 2 days. Complete curing i. e. total resistance to severe environments will be achieved after 7 - 14 days of storage at room temperature depending on film combination and intended use of the laminates.

Storage
When stored in a dry and cool place in the original unopened containers, shelf life is as follows. Once opened containers of LOCTITE® LIOFOL LA 3640 have to be closed tightly again and the material should be consumed within a short period.

Shelf life

LOCTITE® LIOFOL LA 3640, months	9
LOCTITE® LIOFOL LA 6229, months	9

Classification:
Please refer to the corresponding **Safety Data Sheet** for details on:
Hazardous Information
Transport Regulations
Safety Regulations

Additional information

Film requirements
Film additives - particularly slip agent content in LDPE, MDPE and CPP films - packed goods, printing inks, pre-treatment and coating of films are of significant importance to the intended use of the laminates and may affect the performance properties either straight away or later.
Therefore, practical laminating tests as well as checks on the laminates are necessary before going into full scale production.
Further detailed information can be obtained from our "Technical Guide Line for Laminating with Liofol Laminating Adhesives".

Product range
Apart from LOCTITE® LIOFOL LA 3640 / LA 6229 we can offer the following products for the manufacture of film/foil laminates:

- Solventbased, solventless and waterbased laminating adhesives for flexible packaging and special applications
- Adhesives for high gloss lamination
- Primer for extrusion and PVDC-coating
- Coldseal adhesives
- Heatseal coatings
- Special cleaning agents

To obtain product specifications please ask for the relevant technical datasheets.

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